1. Laplace basics (one-sided)

$F(s) = \int_{0^-}^{\infty} f(t)e^{-st}dt$, $s = \sigma + j\omega$. One-sided $\Rightarrow f(t) = 0$ for $t < 0$: always write $f(t)\,u(t)$.
$s = j\omega$: sinusoid; $s = \sigma$: real exponential; $s = \sigma + j\omega$: damped sinusoid. $e^{-st} = e^{-\sigma t}e^{-j\omega t}$: $\sigma > 0$ helps convergence.
ROC: values of s for which the integral converges, e.g. $e^{-at}u(t)$: $\mathrm{Re}(s) > -a$.
FT from LT: $F(\omega) = F(s)|_{s=j\omega}$ *only if* ROC contains the $j\omega$ axis (all poles in open LHP).
Linearity: $\mathcal{L}\{af_1 + bf_2\} = aF_1 + bF_2$. Convolution: $\mathcal{L}\{h * x\} = H(s)X(s)$.

2. Properties/rules NOT on the exam sheet

- Integration: $\mathcal{L}\left\{\int_0^t f(\tau)d\tau\right\} = \dfrac{F(s)}{s}$
- Division by t: $\mathcal{L}\{f(t)/t\} = \int_s^{\infty} F(\nu)\,d\nu$
- Mult. by t^n: $\mathcal{L}\{t^n f(t)\} = (-1)^n \dfrac{d^n F}{ds^n}$ (sheet only has $n = 1$)
- nth derivative: $\mathcal{L}\{f^{(n)}\} = s^n F - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0^-)$. So $\mathcal{L}\{f'''\} = s^3F - s^2f(0^-) - sf'(0^-) - f''(0^-)$
- Delay with step: $\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$. $\mathcal{L}^{-1}\{e^{-as}G\} = g(t-a)u(t-a)$.
- To transform $g(t)u(t-a)$, **rewrite g in terms of $(t-a)$**: $t\,u(t-a) = (t-a)u(t-a) + a\,u(t-a) \to e^{-as}\left(\frac{1}{s^2} + \frac{a}{s}\right)$
- Pulse: $A[u(t-a) - u(t-b)] \to \dfrac{A(e^{-as} - e^{-bs})}{s}$ (e.g. $10[u(t-2) - u(t-3)] \to \frac{10}{s}(e^{-2s} - e^{-3s})$)
- Windowed piece: $f(t)[u(t) - u(t-T)]$ – write f using $(t-T)$ to shift.

3. Extra pairs (all for $t \geq 0$, i.e. $\times u(t)$)

$t^n e^{-at}$	$\dfrac{n!}{(s+a)^{n+1}}$	$\cosh at$	$\dfrac{s}{s^2 - a^2}$
$\cos(\omega t + \phi)$	$\dfrac{s\cos\phi - \omega\sin\phi}{s^2 + \omega^2}$	$\sinh at$	$\dfrac{a}{s^2 - a^2}$
$\sin(\omega t + \phi)$	$\dfrac{s\sin\phi + \omega\cos\phi}{s^2 + \omega^2}$	$t\cos\omega t$	$\dfrac{s^2 - \omega^2}{(s^2 + \omega^2)^2}$
$t\sin\omega t$	$\dfrac{2\omega s}{(s^2 + \omega^2)^2}$	$t^2\sin\omega t$	$\dfrac{\omega(6s^2 - 2\omega^2)}{(s^2 + \omega^2)^3}$
$\delta(t)$	1 $(\delta' \to s)$	ramp kt	k/s^2

$A\cos\omega t + B\sin\omega t = \sqrt{A^2 + B^2}\cos(\omega t - \theta)$, $\theta = \mathrm{atan2}(B, A)$

4. Periodic functions, IVT, FVT

Period T: $F(s) = \dfrac{F_1(s)}{1 - e^{-sT}}$, $F_1 = \mathcal{L}\{f(t)[u(t) - u(t-T)]\}$ = transform of **one period only** (zero-padded; T may include flat gaps).
Ramp kt on $[0, T_0)$: $F_1 = \dfrac{k\left(1 - e^{-sT_0} - sT_0e^{-sT_0}\right)}{s^2}$.
Sawtooth T=1, slope 1: $F = \dfrac{1}{s^2}\dfrac{1 - e^{-s} - se^{-s}}{1 - e^{-s}}$. Slope 2, T=2, ramp on $[0, 1)$: $F = \dfrac{2(1 - e^{-s} - se^{-s})}{s^2(1 - e^{-2s})}$.

IVT: $f(0^+) = \lim_{s\to\infty} sF(s)$ **FVT**: $f(\infty) = \lim_{s\to 0} sF(s)$
$f'(0^+) = \lim_{s\to\infty} s[sF(s) - f(0^+)]$. IVT: F proper (else impulses at t=0). Rel. degree 1 $\Rightarrow f(0^+)$ = ratio of leading coeffs; rel. degree $\geq 2 \Rightarrow f(0^+) = 0$.
FVT valid only if all poles of $sF(s)$ are in the open LHP (a simple pole of F at $s = 0$ is OK). Not valid for undamped oscillation or RHP poles.

Use as a check: step response $V_o = \frac{1}{s}\frac{9}{s^2+2s+10} \Rightarrow v_o(0) = 0$, $v_o(\infty) = 0.9$.

5. Inverse Laplace – recipe

0. If $\deg N \geq \deg D$: divide first. $k \to k\delta(t)$, $s \to \delta'(t)$; e.g. $5 - \frac{2}{s} + \frac{6}{s^2+4} \to 5\delta(t) + (3\sin 2t - 2)u(t)$.
1. Factor $D(s)$, then expand:
Distinct real poles: $F = \sum \frac{A_i}{s-p_i}$, $A_i = (s - p_i)F(s)\big|_{s=p_i} \to \sum A_i e^{p_i t}u(t)$.
Repeated pole $(s+a)^2$: $F = \frac{K_1}{s+a} + \frac{K_2}{(s+a)^2}$, $K_2 = (s+a)^2F\big|_{-a}$, $K_1 = \frac{d}{ds}[(s+a)^2F]\big|_{-a} \to (K_1 + K_2t)e^{-at}$. General r-fold: coeff. of $(s+a)^{-(r-k)} = \frac{1}{k!}\frac{d^k}{ds^k}[(s+a)^rF]_{-a}$.
Complex pair: complete the square $s^2 + bs + c = (s+\alpha)^2 + \omega^2$, $\alpha = \frac{b}{2}$, $\omega^2 = c - \frac{b^2}{4}$:

$$\frac{As + B}{(s+\alpha)^2 + \omega^2} \to e^{-\alpha t}\left[A\cos\omega t + \frac{B - \alpha A}{\omega}\sin\omega t\right]u(t)$$

Residue form: pole $-\alpha + j\omega$ with residue $K \Rightarrow 2|K|e^{-\alpha t}\cos(\omega t + \angle K)$.
Alt.: equate coefficients of s^k after multiplying out (algebraic method).
2. Look up pairs; multiply by $u(t)$; **3.** check with IVT/FVT.
Example: $\frac{s^2+12}{s(s+2)(s+3)} = \frac{2}{s} - \frac{8}{s+2} + \frac{7}{s+3}$.

6. Solving ODEs / zero-input + zero-state

7 steps: ODE $\to$ LT both sides $\to$ insert ICs (0^-) $\to$ solve for $Y(s)$ $\to$ partial fractions $\to$ inverse LT $\to$ check ICs/final value.
For $ay'' + by' + cy = x(t)$:

$$Y(s) = \underbrace{\frac{X(s)}{as^2 + bs + c}}_{\text{zero-state} = H\,X} + \underbrace{\frac{(as + b)y(0^-) + ay'(0^-)}{as^2 + bs + c}}_{\text{zero-input (ICs)}}$$

Integral terms: $\int_0^t y\,d\tau \to Y/s$ (e.g. $y' + 5y + 6\int y = u$: $(s + 5 + \frac{6}{s})Y = \frac{1}{s} + y(0)$).
$u(t) \to \frac{1}{s}$, $t\,u \to \frac{1}{s^2}$, $e^{-at} \to \frac{1}{s+a}$. Solution valid for $t \geq 0$.

7. Circuit elements in the s-domain

R: $V = RI$ $\quad Z = R,\ Y = 1/R$
L: $V = sLI - Li(0^-)$ $\quad Z = sL,\ Y = \frac{1}{sL}$
C: $V = \frac{I}{sC} + \frac{v(0^-)}{s}$ $\quad Z = \frac{1}{sC},\ Y = sC$

L with IC: series sL + voltage source $Li(0^-)$ (so $V = sLI - Li_0$) *or* $sL \parallel$ current source $i(0^-)/s$ in the direction of i ($I = \frac{V}{sL} + \frac{i_0}{s}$).

C with IC: series $\frac{1}{sC}$ + source $\frac{v(0^-)}{s}$ (same polarity as v) *or* $sC \parallel$ current source $Cv(0^-)$ pointing into the + node ($I = sCV - Cv_0$). Zero ICs: $L \to sL$, $C \to 1/sC$ and all DC/AC circuit laws hold with $Z(s)$. Put $s = j\omega$ for sinusoidal steady state.
Series: $Z = \sum Z_k$. Parallel: $Y = \sum Y_k$. Divider: $V_o = V_i \frac{Z_2}{Z_1+Z_2}$; current: $I_1 = I\frac{Z_2}{Z_1+Z_2}$. Also KVL/KCL, nodal, mesh, Thevenin/Norton, superposition.

Procedure: (1) $t < 0$ circuit (DC steady state: L = short, C = open) $\to i_L(0^-), v_C(0^-)$; (2) redraw in s incl. IC sources; (3) solve; (4) partial fractions $\to$ inverse; (5) check IVT/FVT. i_L and v_C are continuous: $(0^+) = (0^-)$. Final DC: L short, C open. Source $A\,u(t) \to A/s$.

8. Standard circuit transfer functions V_o/V_i

RC, out across C (LP) $\quad \frac{1/RC}{s+1/RC} = \frac{a}{s+a}$

C–R, out across R (HP) $\quad \frac{s}{s+1/RC} = \frac{RCs}{1+RCs}$

RL, out across R (LP) $\quad \frac{R/L}{s+R/L}$ $\quad$ across L (HP): $\frac{s}{s+R/L}$

Series RLC, across C (LP) $\quad \frac{1/LC}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$

across R (BP) $\quad \frac{\frac{R}{L}s}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$

across L (HP) $\quad \frac{s^2}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$

Parallel RLC (in: I, out: V) $\quad \frac{V}{I} = \frac{s/C}{s^2 + \frac{s}{RC} + \frac{1}{LC}}$

Voltage divider form $\quad \frac{Z_2}{Z_1 + Z_2}$

Series RLC: $\omega_o = \frac{1}{\sqrt{LC}}$, $\alpha = \frac{R}{2L}$, $\zeta = \frac{\alpha}{\omega_o} = \frac{R}{2}\sqrt{\frac{C}{L}}$, $Q = \frac{\omega_o L}{R} = \frac{1}{\omega_o RC} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{2\zeta}$, $BW = \frac{R}{L} = \frac{\omega_o}{Q}$.

Parallel RLC: $\omega_o = \frac{1}{\sqrt{LC}}$, $\alpha = \frac{1}{2RC}$, $Q = \omega_o RC = R\sqrt{\frac{C}{L}}$, $BW = \frac{1}{RC}$.
Natural response (poles $-\alpha \pm \sqrt{\alpha^2 - \omega_o^2}$):
$\alpha > \omega_o$ overdamped: $K_1 e^{p_1 t} + K_2 e^{p_2 t}$
$\alpha = \omega_o$ critical: $(K_1 + K_2 t)e^{-\alpha t}$
$\alpha < \omega_o$ under: $e^{-\alpha t}(K_1 \cos\omega_d t + K_2 \sin\omega_d t)$, $\omega_d = \sqrt{\omega_o^2 - \alpha^2}$.
Op-amp (inverting): $H = -\frac{Z_f(s)}{Z_{in}(s)}$. LP: $Z_f = R_f \| \frac{1}{sC_f}$, $Z_{in} = R \Rightarrow H = -\frac{R_f}{R}\frac{a}{s+a}$, $a = \frac{1}{R_f C_f}$. With C in parallel with R at input: $H = -\frac{C}{C_f}\frac{s+b}{s+a}$, $b = \frac{1}{RC}$.

9. First-order transient (RC, RL)

$y(t) = y(\infty) + [y(0^+) - y(\infty)]e^{-t/\tau}$, $\tau = RC$ or L/R (R = Thevenin resistance seen by the element).
Step response of $H = \frac{Ka}{s+a}$: $y = K(1 - e^{-at})$, $\tau = \frac{1}{a} = T$, pole $s = -a = -\frac{1}{\tau}$; initial slope $Ka = K/\tau$.

t	τ	2τ	3τ	4τ	5τ
$1 - e^{-t/\tau}$	63.2%	86.5%	95.0%	98.2%	99.3%
$e^{-t/\tau}$	.368	.135	.050	.018	.007

Rise time (10–90%) $= 2.2\tau$; settling (2%) $\approx 4\tau$.
RL step ($v_s = Vu$): $i = \frac{V}{R}(1 - e^{-Rt/L})$ (test: R=100Ω, L=100mH, V=1: $10(1 - e^{-1000t})$ mA). RC: $v_C = V(1 - e^{-t/RC})$.

10. Zero-state responses to common inputs

Impulse: $Y = H \Rightarrow h(t)$. Step: $Y = H/s$. Ramp: $Y = H/s^2$. e^{-bt}: $Y = \frac{H}{s+b}$.
Step response $s(t) = \int_0^t h$, i.e. $h = \frac{ds}{dt}$.
Ex.: $h = e^{-t}u$, $v_i = 10e^{-2t}u \Rightarrow V_o = \frac{10}{(s+1)(s+2)} \to 10(e^{-t} - e^{-2t})u$.
$h = 10e^{-2t}u$, step in $\Rightarrow y = 5(1 - e^{-2t})u$.
Reverse-engineer H: $H = Y/X$ (e.g. $x = 5u$, $y = 0.5\{1 - \sqrt{2}e^{-t}\cos(t - 45°)\} \Rightarrow H = \frac{0.2}{s^2+2s+2}$: expand $\sqrt{2}\cos(t - 45°) = \cos t + \sin t$).
Initial conditions: $Y = HX + \frac{\text{IC terms}}{D(s)}$ (see §6).

11. Transfer function $H(s) = Y(s)/X(s)$ (zero ICs)

$Y = HX$; $h(t) = \mathcal{L}^{-1}\{H\}$; $y = h * x = \int_0^t h(\tau)x(t-\tau)d\tau$ (causal LTI). $H(\omega) = H(s)|_{s=j\omega}$.

Types: V_o/V_i voltage gain, I_o/I_i current gain (dimensionless); V/I impedance (Ω); I/V admittance (S). One circuit has many H.

ODE $\leftrightarrow$ H: $\sum_k a_k y^{(k)} = \sum_k b_k x^{(k)} \iff H = \dfrac{\sum b_k s^k}{\sum a_k s^k}$ (cross-multiply, $s^k \leftrightarrow d^k/dt^k$). Order $n = \deg D$; characteristic eq. $D(s) = 0$.

Ex.: $K = 3$, zero -4, poles $-1 \pm j2$: $H = \frac{3(s+4)}{s^2+2s+5} \Rightarrow y'' + 2y' + 5y = 3x' + 12x$.

Cascade (isolated stages) $H = H_1 H_2$; **parallel** $H = H_3 + H_4$. E.g. $\frac{4s+28}{s^2+6s+5} = \frac{4s+28}{s+1} \cdot \frac{1}{s+5} = \frac{6}{s+1} - \frac{2}{s+5}$. Cascade $\leftrightarrow$ factorise; parallel $\leftrightarrow$ partial fractions.

		$H(s)$	$\lvert H(j\omega)\rvert$	$\angle H$
Ideal blocks:	delay T	e^{-sT}	1	$-\omega T$
	differentiator	s	ω	$+90°$
	integrator	$1/s$	$1/\omega$	$-90°$

Delay: $y(t) = x(t-T) \Rightarrow Y = e^{-sT}X$.

12. Poles & zeros

$H = \dfrac{P(s)}{Q(s)} = K\dfrac{\prod(s-z_i)}{\prod(s-p_i)}$. Zeros: $P = 0$ ($H = 0$); poles: $Q = 0$ ($H \to \infty$). Plot $\circ$ zeros, $\times$ poles; real systems $\Rightarrow$ complex roots in conjugate pairs. $s^2 + bs + c = 0$: $s = -\frac{b}{2} \pm \sqrt{\frac{b^2}{4} - c}$ (complex if $c > \frac{b^2}{4}$).

Find K from a known value: $H(s_0)$ given $\Rightarrow K = H(s_0)\frac{\prod(s_0-p_i)}{\prod(s_0-z_i)}$ (e.g. $H(0)$ is the DC gain).

Pole at origin = integrator; zero at origin blocks DC.

13. Pole location $\Rightarrow$ response, stability

Real LHP $-a$	e^{-at} decays, $\tau = 1/a$; farther from $j\omega$ = faster
Real RHP $+a$	e^{at} grows: **unstable**
Origin	constant (step); double: ramp t (unstable)
$\pm j\omega_a$	sustained oscillation, **not BIBO**
$-\sigma \pm j\omega_a$	$e^{-\sigma t}\sin(\omega_a t + \phi)$: σ = decay rate, ω_a = oscillation freq.
$+\sigma \pm j\omega_a$	growing oscillation (unstable)
Repeated LHP	$t^k e^{-at}$ decays

$h = \frac{1}{\omega_a}e^{\sigma_a t}\sin\omega_a t$ for $H = \frac{1}{(s-p_1)(s-p_2)}$, $p_{1,2} = \sigma_a \pm j\omega_a$. Fixed ω_a, more negative σ: same oscillation, faster decay. Fixed σ, larger ω_a: same envelope, faster oscillation.

BIBO stable $\iff \int|h| < \infty \iff$ all poles in **open** LHP ($\mathrm{Re}\, p_i < 0$) (causal, rational). Zeros do not affect stability. Causal + stable $\Rightarrow$ all poles in LHP. A causal system with a RHP/imag. pole cannot be stable (ex.: $\frac{3s-1}{(s+3)(s-2)} \to h = (2e^{-3t} + e^{2t})u(t)$ grows: unstable).

14. Second-order system $H = \dfrac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \dfrac{a}{s^2 + bs + a}$

$\omega_n^2 = a$, $2\zeta\omega_n = b$, $Q = \frac{1}{2\zeta}$. DC gain 1. Unit step $y = \mathcal{L}^{-1}\{H/s\}$:

$\zeta > 1$ over	poles $p_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2-1}$, $y = 1 + \dfrac{p_2 e^{p_1 t} - p_1 e^{p_2 t}}{p_1 - p_2}$ (no overshoot)
$\zeta = 1$ crit.	double pole $-\omega_n$; $y = 1 - (1+\omega_n t)e^{-\omega_n t}$
$0 < \zeta < 1$ under	$-\sigma \pm j\omega_d$, $\sigma = \zeta\omega_n$, $\omega_d = \omega_n\sqrt{1-\zeta^2}$; $y = 1 - e^{-\sigma t}\left[\cos\omega_d t + \frac{\sigma}{\omega_d}\sin\omega_d t\right] = 1 - \frac{\omega_n}{\omega_d}e^{-\sigma t}\sin(\omega_d t + \cos^{-1}\zeta)$
$\zeta = 0$ undamped	$\pm j\omega_n$; $y = 1 - \cos\omega_n t$

Poles (underdamped) lie on circle radius ω_n, with $\cos\theta = \zeta$ (θ from negative real axis). $\%OS = 100e^{-\pi\zeta/\sqrt{1-\zeta^2}}$, peak time $t_p = \pi/\omega_d$, $t_s(2\%) \approx 4/\sigma$. E.g. $\frac{9}{s^2+2s+9}$: $\omega_n = 3, \zeta = \frac{1}{3}, \sigma = 1, \omega_d = \sqrt{8}$. Overdamped: **dominant pole** = closest to $j\omega$ axis sets speed.

Impulse resp. $\frac{1}{s^2+2s+10}$-type: $h = \frac{1}{\omega}e^{-\sigma t}\sin\omega t$. **Sketching** $h(t)$: $h(0^+) = \lim sH$ (0 if rel. deg. ≥ 2), envelope $\pm Ae^{-\sigma t}$, oscillation freq ω_a (period $2\pi/\omega_a$), zero crossings every π/ω_a. $H = \frac{9}{s^2+2s+10} \Rightarrow h = 3e^{-t}\sin 3t\, u(t)$: starts at 0, envelope $\pm 3e^{-t}$.

15. Frequency response $H(j\omega) = |H|e^{j\phi}$

Stable system, $x = A\cos(\omega t + \theta) \Rightarrow y_{ss} = A\,|H(j\omega)|\cos(\omega t + \theta + \angle H(j\omega))$.

Real system: $|H(-j\omega)|$ even, $\angle H(-j\omega)$ odd.

Vector method: $|H(j\omega)| = |K|\dfrac{\prod|j\omega - z_i|}{\prod|j\omega - p_i|}$, $\angle H = \angle K + \sum\angle(j\omega - z_i) - \sum\angle(j\omega - p_i)$. Vector from pole/zero to the point $j\omega$.

Pole near $j\omega$ axis $\Rightarrow$ peak near $\omega \approx \mathrm{Im}\, p$; zero near/on axis $\Rightarrow$ dip/null (zero on axis at $\omega_z \Rightarrow |H| = 0$).

LHP zero $-a$: $\angle(j\omega + a) = \tan^{-1}\frac{\omega}{a}$ ($0 \to 90°$); RHP zero $+a$: $\angle(j\omega - a) = 180° - \tan^{-1}\frac{\omega}{a}$; same $|H|$ as the LHP mirror.

Eg. $\frac{s}{(s+1)^2+10^2}$: $|H(j10)| = \frac{10}{|1|\cdot|1+j20|} = \frac{10}{\sqrt{401}}$, $|H(0)| = 0$, $|H(\infty)| \to 0$.

Sketch checklist: (i) DC: $H(0)$; (ii) $\omega \to \infty$: leading terms; (iii) peaks/nulls from nearby poles/zeros; (iv) even symmetry for $\omega < 0$.

Slopes: far above all corners: $-20(n-m)$ dB/dec (n poles, m zeros); k zeros at origin: $+20k$ dB/dec at low ω. Each real pole/zero corner changes slope by $\mp 20/\pm 20$ dB/dec; phase of $\frac{1}{1+s/\omega_c}$: 0 ($\omega \ll \omega_c$), $-45°$ (ω_c), $-90°$ ($\omega \gg \omega_c$).

dB: $20\log_{10}|H|$ (V, I gain); $10\log_{10}$ power ratios. $10^{x/20}$.

ratio	0.1	0.5	$\frac{1}{\sqrt{2}}$	1	$\sqrt{2}$	2	10	100
dB	-20	-6	-3	0	3	6	20	40

$\log_{10} 2 = 0.301$, 20 dB/dec = 6 dB/oct. Power: $10\log_{10} 2 = 3$dB, $10\log_{10}\frac{1}{2} = -3$dB.

Cutoff/bandwidth: $|H(j\omega_c)| = H_{max}/\sqrt{2}$ (-3 dB, half power). LPF: $BW = \omega_c$. BPF: $BW = \omega_u - \omega_l$.

16. First-order filters

	LP	HP
$H(s)$	$\frac{\omega_c}{s+\omega_c}$	$\frac{s}{s+\omega_c}$
$\lvert H\rvert$	$\frac{1}{\sqrt{1+(\omega/\omega_c)^2}}$	$\frac{\omega/\omega_c}{\sqrt{1+(\omega/\omega_c)^2}}$
$\angle H$	$-\tan^{-1}\frac{\omega}{\omega_c}$	$90^\circ-\tan^{-1}\frac{\omega}{\omega_c}$
at ω_c	0.707, −3dB, −45°	0.707, −3dB, +45°
DC/HF	1 / 0	0 (blocks DC) / 1
slope	−20 dB/dec above ω_c	+20 dB/dec below ω_c
RC	$\omega_c=\frac{1}{RC}$ (out on C)	$\omega_c=\frac{1}{RC}$ (out on R)
RL	$\omega_c=\frac{R}{L}$ (out on R)	$\omega_c=\frac{R}{L}$ (out on L)

$f_c=\frac{\omega_c}{2\pi}=\frac{1}{2\pi RC}$. *Design*: $C=\frac{1}{2\pi Rf_c}$ (R = 1.6kΩ, 1 kHz $\Rightarrow C=99.5$nF). Check $|H|=\frac{1/RC}{\sqrt{\omega^2+(1/RC)^2}}$.

DC gain $\neq 1$: $H=\frac{K\omega_c}{s+\omega_c}$, cutoff still ω_c, passband $=K$.

Cascade of n identical 1st-order LPs: $H=\left(\frac{\omega_c}{s+\omega_c}\right)^n$, slope $-20n$ dB/dec, −3dB point at $\omega=\omega_c\sqrt{2^{1/n}-1}$ (n=2: $0.644\omega_c$; at ω_c it is −6 dB).

17. Second-order filters ($b_o=\omega_o/Q$, ω_o = resonant/centre freq.)

Poles of $s^2+\frac{\omega_o}{Q}s+\omega_o^2$: $s=-\frac{\omega_o}{2Q}\pm j\omega_o\sqrt{1-\frac{1}{4Q^2}}$. All on circle $|s|=\omega_o$ for $Q\geq\frac{1}{2}$; raising Q moves them toward $j\omega$ axis.
$Q>\frac{1}{2}$: complex; $Q=\frac{1}{2}$: repeated real ($\zeta=1$); $Q<\frac{1}{2}$: real distinct (cutoff set by dominant pole α_1, the one nearest $j\omega$).

LP: $H=\frac{\omega_o^2}{s^2+\frac{\omega_o}{Q}s+\omega_o^2}$, $|H|=\frac{\omega_o^2}{\sqrt{(\omega_o^2-\omega^2)^2+(\frac{\omega\omega_o}{Q})^2}}$

DC gain 1; $|H(j\omega_o)|=Q$, $\angle=-90^\circ$; $\omega\to\infty$: −40 dB/dec, $\angle\to-180^\circ$.

Peak if $Q>\frac{1}{\sqrt{2}}$: $\omega_r=\omega_o\sqrt{1-\frac{1}{2Q^2}}$, $|H|_{max}=\frac{Q}{\sqrt{1-\frac{1}{4Q^2}}}$. $Q=\frac{1}{\sqrt{2}}$: maximally flat, −3dB at ω_o.

HP: $H=\frac{s^2}{s^2+\frac{\omega_o}{Q}s+\omega_o^2}$: $|H(j\omega_o)|=Q$, $\angle$: $180^\circ\to0^\circ$, +40 dB/dec below ω_o; 2 zeros at origin.

BP: $H=\frac{\frac{\omega_o}{Q}s}{s^2+\frac{\omega_o}{Q}s+\omega_o^2}=\frac{b_os}{s^2+b_os+\omega_o^2}$; zero at origin.

$|H|=\frac{\omega\omega_o/Q}{\sqrt{(\omega_o^2-\omega^2)^2+(\omega\omega_o/Q)^2}}$, $|H(j\omega_o)|=1$ (max), $\angle=0^\circ$ at ω_o; $\angle H=90^\circ-\text{Arg}[(\omega_o^2-\omega^2)+j\frac{\omega\omega_o}{Q}]$ ($+90^\circ\to-90^\circ$).

$H=\frac{Ks}{s^2+bs+\omega_o^2}\Rightarrow$ peak gain K/b at $\omega=\omega_o$ (e.g. $\frac{s}{s^2+2s+101}$: $b=2$, $\omega_o=\sqrt{101}$, peak $\frac{1}{2}$).

Centre $\omega_o=\sqrt{\omega_l\omega_u}$ (geometric mean; where H is purely real). $BW=\omega_u-\omega_l=\omega_o/Q$. $Q=\frac{\omega_o}{BW}$.

$\omega_{u,l}=\omega_o\left[\sqrt{1+\frac{1}{4Q^2}}\pm\frac{1}{2Q}\right]$; cutoffs at $|H|=\frac{1}{\sqrt{2}}H_{max}$. Large Q: $\omega_{u,l}\approx\omega_o\pm\frac{BW}{2}$.

Series RLC (out on R): $\omega_o=\frac{1}{\sqrt{LC}}$, $BW=\frac{R}{L}$, $Q=\frac{\omega_oL}{R}$. Resonance: $j\omega_oL+\frac{1}{j\omega_oC}=0$.

Design from poles $-a\pm j\omega_d$: $\omega_o=\sqrt{a^2+\omega_d^2}$, $b=2a$, $Q=\frac{\omega_o}{2a}$.
BPF with centre 10, zero at 0 and ∞: $H=\frac{Ks}{(s+a)^2+10^2}$; small $a\Rightarrow$ narrow/sharp (high Q), large $a\Rightarrow$ wide, flat.

Notch: $H=\frac{s^2+\omega_z^2}{s^2+b_ps+\omega_p^2}$. Standard: $\omega_p=\omega_z=\omega_o$: $H=\frac{s^2+\omega_o^2}{s^2+\frac{\omega_o}{Q}s+\omega_o^2}$, $|H|=\frac{|\omega_o^2-\omega^2|}{\sqrt{(\omega_o^2-\omega^2)^2+(\omega\omega_o/Q)^2}}$

$|H(0)|=|H(\infty)|=1$, $|H(j\omega_o)|=0$; larger $Q_p\Rightarrow$ narrower notch ($BW=\omega_o/Q$). Pure notch (no poles): $\frac{s^2+\omega_z^2}{\omega_z^2}$ rises as ω^2.

Pole form with angle θ: $s^2+2\omega_o\cos\theta\,s+\omega_o^2$: $\zeta=\cos\theta$, $Q=\frac{1}{2\cos\theta}$; $\theta=60^\circ\Rightarrow Q=1$, $b=\omega_o$. 50 Hz hum: $\omega_o=2\pi\cdot50=314.2$ rad/s.

18. Identities & quick numerics

$e^{j\theta}=\cos\theta+j\sin\theta$; $\cos\theta=\frac{e^{j\theta}+e^{-j\theta}}{2}$, $\sin\theta=\frac{e^{j\theta}-e^{-j\theta}}{2j}$.
$|a+jb|=\sqrt{a^2+b^2}$, $\angle=\text{atan2}(b,a)$; $|z_1z_2|=|z_1||z_2|$, $\angle(z_1/z_2)=\angle z_1-\angle z_2$.
$\tan^{-1}$: $1\to45^\circ$, $\frac{1}{\sqrt{3}}\to30^\circ$, $\sqrt{3}\to60^\circ$. $\frac{1}{\sqrt{2}}=0.7071$, $\sqrt{2}=1.414$, $\sqrt{3}=1.732$, $\sqrt{5}=2.236$.
$\omega=2\pi f$; $\pi=3.1416$. $(a+jb)(a-jb)=a^2+b^2$.

Common slips: forgetting $u(t)$ or the 0^- ICs; mixing $0^-/0^+$; FVT on unstable/oscillating F; 20 log vs 10 log; f vs ω (2π); phase quadrant (use Arg of the whole complex denominator); sign of IC sources; applying cascade rule when stages load each other; partial fractions on improper F (divide first).